AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

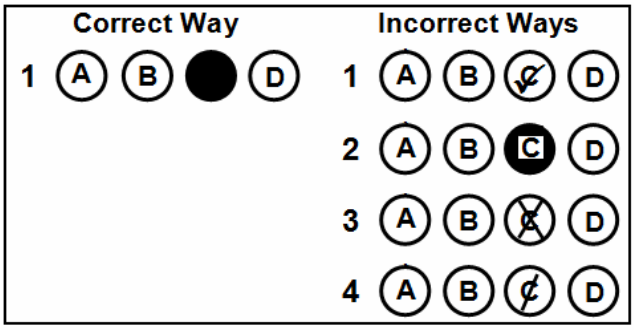
MAY 2015

General Mathematics Paper I

Time: 40 minutes Marks: 30

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. If $2(v+w)=12$ and $w=2,$ then v is equal to

   A. 8
   B. 6
   C. 4
   D. 3

2. The expression $\dfrac{(a+b)^3}{a^2-b^2}$ in the simplest form is equal to

   A. $\dfrac{(a+b)^2}{a-b}$

   B. $\dfrac{(a-b)^2}{a+b}$

   C. $\dfrac{(a+b)^2}{a+b}$

   D. $\dfrac{(a-b)^2}{a-b}$

3. $\left(\sqrt{a}+\sqrt{b}\right)\left(\sqrt{a}-\sqrt{b}\right)$ is equal to

   A. $2\sqrt{a}$

   B. $-2\sqrt{b}$

   C. $a-b$

   D. a^2-b^2

4. The factors of x^4-y^4 are

   A. $(x+y)^3(x-y)$

   B. $(x+y)^2(x-y)^2$

   C. $(x+y)(x-y)\left(x^2-y^2\right)$

   D. $(x+y)(x-y)\left(x^2+y^2\right)$

5. When x^3-2 is divided by $x+1,$ the remainder is

   A. 3
   B. 1
   C. -1
   D. -3

6. Which one of the following is a factor of $x^3 - x^2 + 2$?

A. $x - 1$
B. $x + 1$
C. $x - 2$
D. $x + 2$

7. $\sqrt{a^2 + \frac{1}{a^2} - 2}$ is equal to

A. $\left(a + \frac{1}{a}\right)$

B. $\left(a - \frac{1}{a}\right)$

C. $\left(a^2 + \frac{1}{a^2}\right)$

D. $\left(a^2 - \frac{1}{a^2}\right)$

8. The least common multiple of $\left(x^3 - 1\right)$ and $\left(x^2 + x + 1\right)$ is

A. $x - 1$

B. $x^2 + x + 1$

C. $(x-1)\left(x^2 + x + 1\right)$

D. $(x-1)\left(x^2 + x + 1\right)^2$

9. The highest common factor of $25x$, $5x^2$ and $20x^3$ is

A. $5x$
B. $5x^2$
C. $25x^2$
D. $20x^3$

10. If $p < q < 0$, where p and q are real numbers, then which of the following is TRUE?

A. $pq < 0$
B. $pq > 0$
C. $pq \leq 0$
D. $pq \geq 0$

11. If $\sqrt{\frac{5-2x}{2}} = 0$, then the value of x is equal to

A. $\frac{5}{4}$

B. $\frac{5}{2}$

C. $\frac{4}{5}$

D. $\frac{2}{5}$

12. If $|2x-3| = 1$, then the solution set is

A. $\{1\}$

B. $\{2\}$

C. $\{1, 2\}$

D. $\{-1, -2\}$

13. If one root of $2x^2 - x = 0$ is 0, the other root is

A. 2

B. $\frac{1}{2}$

C. $-\frac{1}{2}$

D. -2

14. What should be added to $x^2 + 5x$ to make it a perfect square?

A. 5

B. 25

C. $\frac{5}{2}$

D. $\frac{25}{4}$

15. The additive inverse of the matrix $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ is

A. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

B. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

C. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

D. $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

16. If $\begin{bmatrix} 2 & x \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}$, then the value of x is

A. 2
B. 0
C. -1
D. -2

17. The multiplicative inverse of the matrix $\begin{bmatrix} 6 & 3 \\ -y & 1 \end{bmatrix}$ is NOT possible if

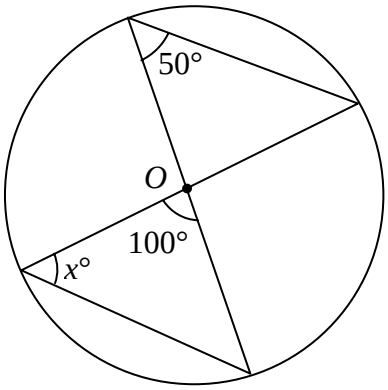
A. $y = 0$
B. $y = 2$
C. $y = -2$
D. $y = -6$

18. If P is a matrix of order 2×3 and Q is a matrix of order 3×1, then the order of the matrix $P \times Q$ is

A. 2×1
B. 2×3
C. 3×2
D. 3×3

19. In the given circle with centre O, the value of x is

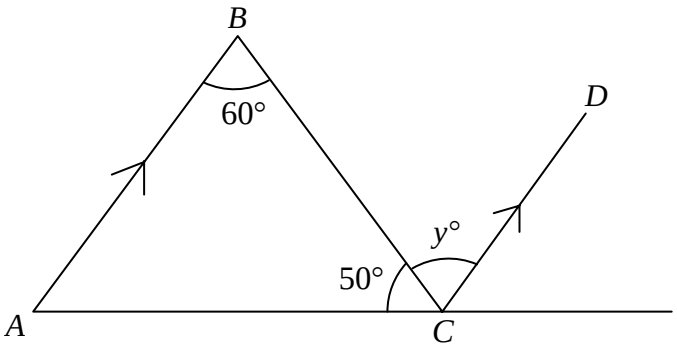
A. 30°
B. 40°
C. 50°
D. 80°



NOT TO SCALE

20. In the given figure, ABC is a triangle and $\overline{AB}$ is parallel to $\overline{CD}$. The value of y is

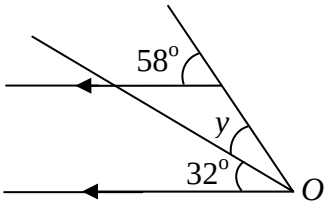
A. 50°
B. 60°
C. 70°
D. 90°



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21. In the figure, the value of y is

A. 13°
B. 26°
C. 29°
D. 32°



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22. If two or more angles combine together to form a straight line, the sum of their measure is

A. 60°
B. 90°
C. 180°
D. 360°

23. $\overline{AC}$ is one of the medians of the given triangle *ABD*. Therefore, $m\overline{BC}$ will be equal to

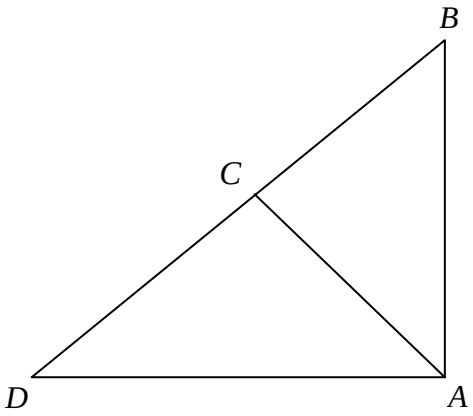
A. $m\overline{CD}$

B. $m\overline{AB}$

C. $m\overline{AC}$

D. $m\overline{AD}$

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24. Which of the following set of values is possible for the given triangle?

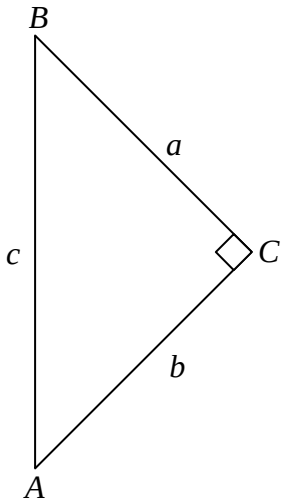
A. $a = 4$, $b = 5$ and $c = 3$

B. $a = 5$, $b = 3$ and $c = 4$

C. $a = 13$, $b = 5$ and $c = 12$

D. $a = 5$, $b = 12$ and $c = 13$

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25. The area of a cube is 4cm^2. Its volume is

A. $8\ \text{cm}^3$

B. $16\ \text{cm}^3$

C. $36\ \text{cm}^3$

D. $64\ \text{cm}^3$

26. The area of the given parallelogram *ABCD* is

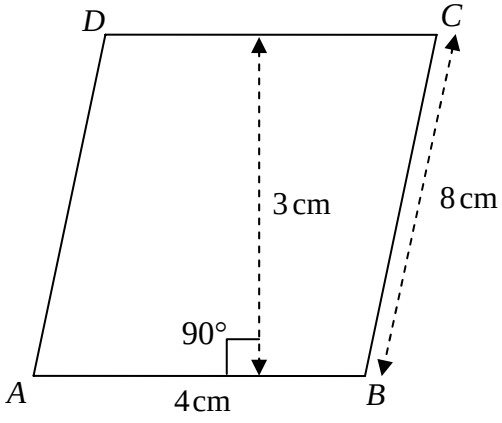
A. 6cm^2

B. $12\,\text{cm}^2$

C. 24cm^2

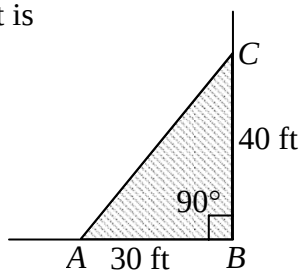
D. $32\,\text{cm}^2$

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27. The figure represents a triangular plot *ABC*. The area of the plot is

A. $2500\,\text{ft}^2$
B. $1200\,\text{ft}^2$
C. $600\,\text{ft}^2$
D. $120\,\text{ft}^2$



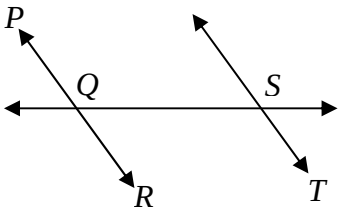
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28. The point collinear to the points (0, 0) and (1, 1) is

A. (2, 2)
B. (2, 0)
C. (0, 1)
D. (1, – 1)

29. In the given figure, three collinear points are *Q*,

A. *S* and *T*.
B. *P* and *S*.
C. *R* and *S*.
D. *P* and *R*.

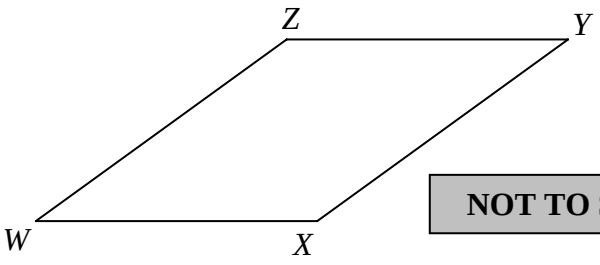


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30. Which of the following is/ are TRUE for the given parallelogram?

I. Point *W*, *Y* and *Z* are collinear.
II. Point *W*, *X* and *Z* may form a triangle.
III. $m\overline{WX} = m\overline{ZY}$

A. II only
B. I and II
C. I and III
D. II and III



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END OF PAPER